RGIPT

Subject: Artificial Intelligence

Class: 12

Time Allowed: 90 minutes **Maximum Marks: 22**

All questions are compulsory unless stated otherwise.

Name: ___________________________

Roll Number: ____________________

Class: 12 Section: ________

Section A: Assertion-Reason Questions

Answer each assertion-reason pair. Each question carries 2 marks.

1. [Easy] Assertion (A): Activation functions are required to introduce non-linearity in neural networks. Reason (R): Without activation functions, stacking multiple linear layers is equivalent to a single linear transformation. [2 Marks]

2. [Easy] Assertion (A): The mean squared error loss penalizes larger errors more than smaller errors. Reason (R): MSE squares the difference between prediction and target. [2 Marks]

3. [Moderate] Assertion (A): Gradient descent always finds the global minimum of the loss surface. Reason (R): The loss surface of neural networks is convex. [2 Marks]

4. [Easy] Assertion (A): Bias term in a neuron allows the activation function to shift left or right. Reason (R): Bias adds a constant to the weighted sum before activation. [2 Marks]

Section B: True / False Questions

Mark each statement as True or False. Each question carries 1 mark.

1. [Easy] ReLU activation outputs negative values as zero. [1 Marks]

2. [Easy] In backpropagation, weights are updated before the loss is computed. [1 Marks]

3. [Easy] Learning rate determines the size of steps taken during gradient descent. [1 Marks]

Section C: Short Answer Questions

Answer briefly. Each question carries 3 marks.

1. [Moderate] Explain why stacking only linear layers without activation functions does not increase the expressive power of a neural network. [3 Marks]

2. [Moderate] List and briefly describe the four steps of the training loop in neural networks. [3 Marks]

Section D: Long Answer Question

Answer in detail. This question carries 5 marks.

1. [Challenging] Discuss the role of backpropagation and gradient descent in training a neural network, including how errors are propagated and weights are updated. [5 Marks]

End of Question Paper

Answer Key:

1. Both A and R are true and R correctly explains A
2. Both A and R are true and R correctly explains A
3. Both A and R are false
4. Both A and R are true and R correctly explains A
5. True
6. False
7. True
8. Because the composition of linear transformations is itself a linear transformation; mathematically, multiple linear layers collapse into a single matrix multiplication plus bias, so no new non-linear decision boundaries can be formed.
9. 1. Forward Pass – data moves through the network to produce a prediction. 2. Calculate Loss – compare the prediction with the true label to obtain a scalar error. 3. Backpropagate – compute gradients of the loss w.r.t. each weight by propagating error backward. 4. Update Weights – adjust each weight using gradient descent (weight = weight – learning_rate × gradient).
10. Backpropagation is the algorithm that computes how much each weight contributed to the overall error. Starting from the loss at the output, the chain rule is applied layer by layer to obtain gradients of the loss with respect to every weight and bias. These gradients indicate the direction and magnitude of change needed to reduce the error. Gradient descent then uses these gradients to update the parameters: weight_new = weight_old – η × gradient, where η is the learning rate. By repeatedly performing forward passes, loss calculation, backpropagation, and gradient-descent updates, the network iteratively moves toward a set of parameters that minimize the loss, ideally reaching a (global) minimum of the loss surface.